

Indian Institute of Technology Guwahati

Department of Electronics and Electrical Engineering

Design and Analysis of Two-Stage OTA

in TSMC 180nm Technology

Course Assignment

EE206 – Analog Circuits

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Course: EE206 – Analog Circuits

Technology Node: TSMC 180nm CMOS

1. Design Objective and Specifications

The objective of this project is to design a two-stage Operational Transconductance Amplifier (OTA) utilizing the TSMC 0.18 μm CMOS process. The amplifier is designed to operate in a closed-loop non-inverting configuration with a voltage gain of 2, driving a generic load capacitance.

The primary design constraints and technology parameters are as follows:

- **Technology:** TSMC 180 nm ($V_{Tn} = 0.37$ V, $V_{Tp} = 0.39$ V)
- **Mobility:** $\mu_n C_{ox} = 230$ $\mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 100$ $\mu\text{A}/\text{V}^2$
- **Supply Voltage (V_{DD}):** 1.8 V
- **DC Open-Loop Gain:** ≥ 40 dB
- **Bias Network:** Reference current is exactly $1/10^{\text{th}}$ of the first-stage tail current.
- **Transient Requirement:** Stable response to a 0.2V input step with 0.1ns rise/fall times.

2. Theoretical Hand Calculations

To mitigate channel-length modulation effects and ensure the intrinsic output resistance (r_o) remains high enough to achieve the > 40 dB gain specification, a non-minimum channel length of $L = 0.5\mu\text{m}$ was selected for all transistors. The initial overdrive voltage was assumed to be $V_{OV} = 200$ mV.

The tail current for the differential pair was set to $I_{tail} = 100\mu\text{A}$, effectively setting the reference current to $10\mu\text{A}$ to satisfy the 1:10 mirror constraint. The tail current splits evenly, providing $50\mu\text{A}$ to each branch of the differential pair.

Transistor aspect ratios were derived using the standard square-law model for saturation:

$$W = L \cdot \frac{2I_D}{\mu C_{ox} V_{OV}^2} \quad (1)$$

For a V_{OV} of 0.2V, the required theoretical widths are:

- **Bias Reference (M7):** $I_D = 10\mu\text{A} \implies W_7 \approx 1.1\mu\text{m}$
- **Tail Source (M0):** $I_D = 100\mu\text{A} \implies W_0 \approx 10.9\mu\text{m}$
- **Differential Pair (M1, M2):** $I_D = 50\mu\text{A} \implies W_{1,2} \approx 5.4\mu\text{m}$
- **PMOS Active Load (M3, M4):** $I_D = 50\mu\text{A} \implies W_{3,4} = 12.5\mu\text{m}$

Device	Type	Target I_D (μA)	Theoretical W (μm)
M7	NMOS (Bias)	10	1.1
M0	NMOS (Tail)	100	10.9
M1, M2	NMOS (Diff Pair)	50	5.4
M3, M4	PMOS (Active Load)	50	12.5
M5	PMOS (2nd Stage Amp)	100	25.0
M6	NMOS (2nd Stage Bias)	100	10.9

Table 1: Theoretical hand-calculated transistor widths ($L = 0.5\mu\text{m}$, $V_{OV} = 0.2\text{V}$)

Two distinct architectures were explored for the second stage to evaluate the trade-off between power efficiency and high-speed driving capability.

3. Design 1: Power-Optimized Architecture

The first architecture matches the current of the second stage to the tail current of the first stage (a 1:1 ratio, $I_{D5,6} \approx 100\mu\text{A}$). This minimizes overall static power dissipation but requires precise frequency compensation.

3.1. Circuit Sizing and DC Operating Point

To eliminate systematic offset, the current density of M5 was matched to M3. Slight manual tuning was performed in LTspice to perfectly center the output node voltage, yielding the final dimensions and exact operating currents listed below.

Device	Type	W/L (μm)	Sim. I_D (μA)	$ V_{GS} $ (V)	$ V_{GS} - V_T $ (V)	g_m (μS)	r_o ($\text{k}\Omega$)
M7	NMOS	1.1/0.5	10.00	0.624	0.254	78.7	1000
M0	NMOS	13.0/0.5	100.80	0.624	0.254	793.7	100
M1, M2	NMOS	6.0/0.5	50.40	0.684	0.314	321.0	200
M3, M4	PMOS	13.0/0.5	50.40	0.685	0.295	341.7	200
M5	PMOS	25.6/0.5	100.57	0.723	0.333	604.0	100
M6	NMOS	11.6/0.5	100.57	0.624	0.254	791.9	100

Table 2: Simulated operating points and estimated small-signal parameters for Design 1. (*Calculated from exact LTspice node voltages; r_o assumes $\lambda \approx 0.1 \text{ V}^{-1}$*)

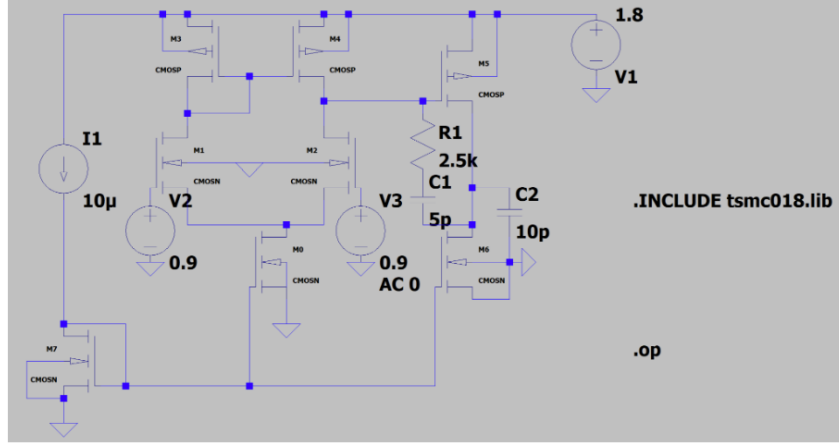


Figure 1: DC Operating Point Schematic for Design 1 ($100\mu\text{A}$ second stage).

3.2. Compensation and AC Analysis (Bode Plot)

Because the second stage operates at a low current ($100\mu\text{A}$), its transconductance (g_{m5}) is modest. Using a standard Miller capacitor alone introduces a Right-Half Plane (RHP) zero at a low enough frequency to severely degrade the phase margin.

To counteract this, a series nulling resistor ($R_1 = 2.5\text{ k}\Omega$) was placed in series with the Miller capacitor ($C_1 = 5\text{ pF}$). This effectively pushes the RHP zero into the left-half plane. A capacitive load of $C_2 = 10\text{ pF}$ was included to model realistic driving conditions.

The amplifier easily clears the 40 dB constraint. The AC analysis yields a DC open-loop gain magnitude of **68.28 dB**. The nulling resistor successfully stabilizes the circuit, resulting in a phase margin of **67.5°** and a Gain Bandwidth Product (GBW) of **12.803 MHz**.

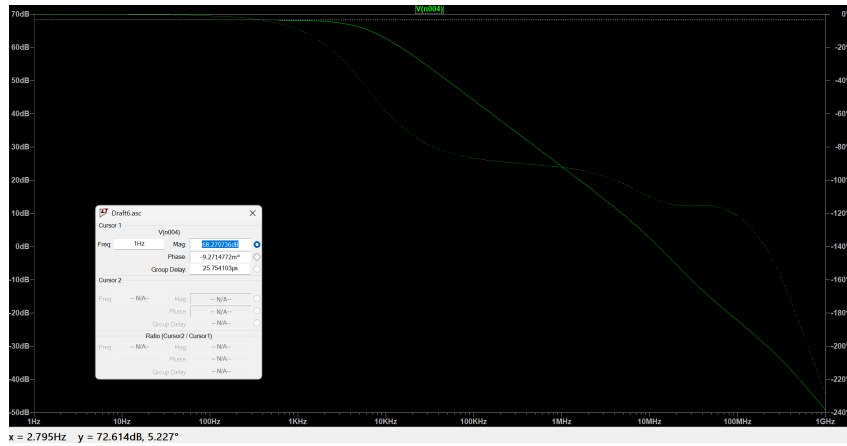


Figure 2: Bode Plot - DC gain = 68.28 dB for Design 1



Figure 3: Bode Plot - Gain Bandwidth Product (0 dB) = 12.803 MHz for Design 1

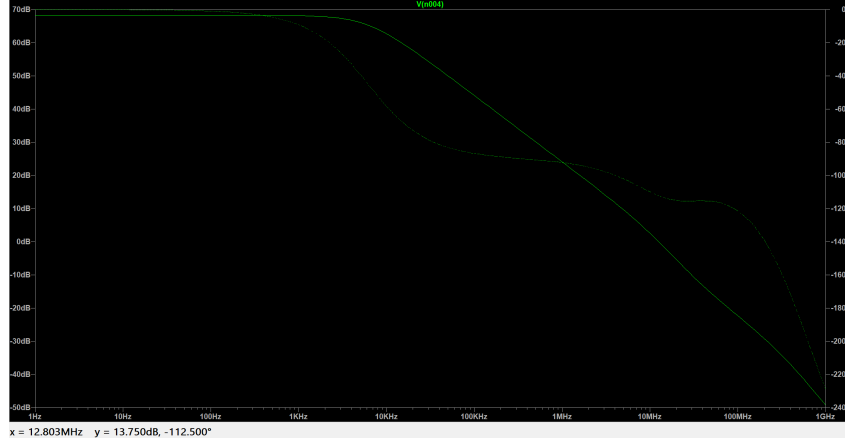


Figure 4: Bode Plot - Phase Margin = 67.5° for Design 1

3.3. Transient Response and Accuracy

The OTA was wired in a closed-loop configuration with feedback resistors $R_f = 10 \text{ k}\Omega$ and $R_{in} = 10 \text{ k}\Omega$ to provide an ideal closed-loop gain of 2. Applying a 0.2V input step (0.9V to 1.1V, configured as PULSE(0.9 1.1 0 0.1n 0.1n 10u 20u)) resulted in an output step change.

- **Target Output Step:** 0.4000 V
- **Measured Output Step:** 0.3990 V
- **Steady-state gain error:** 0.251%

This confirms highly stable closed-loop operation with excellent gain accuracy and minimal ringing.

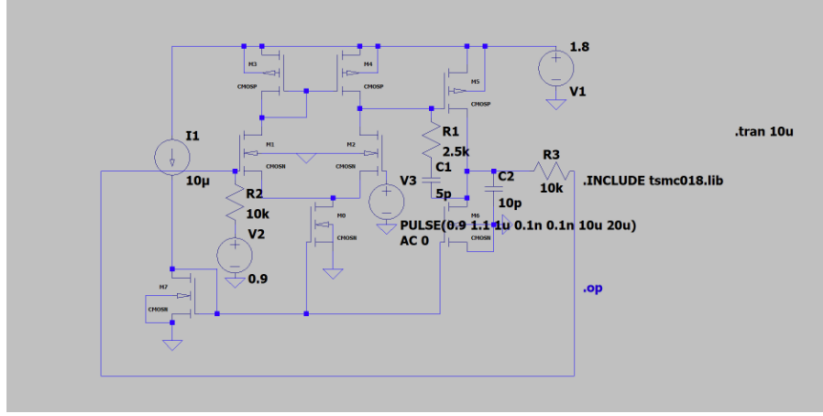


Figure 5: Closed-loop circuit - Non-Inverting Amplifier of gain = 2 using Design 1.

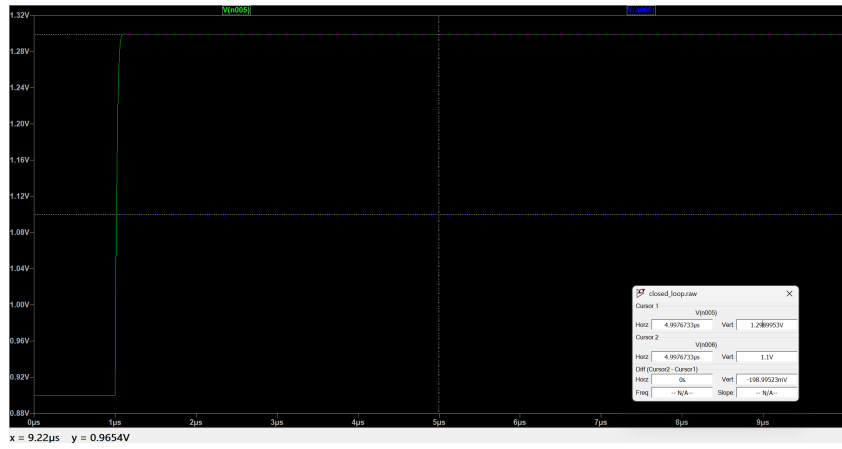


Figure 6: Closed-loop transient step response for Design 1.

3.4. Power Dissipation

The total static power dissipation is derived from the steady-state currents and the 1.8V supply. The exact per-device power dissipation is:

- **First Stage & Bias:** $M0 = 22.21\mu\text{W}$, $M1/M2 = 45.17\mu\text{W}$ (total), $M3/M4 = 34.44\mu\text{W}$ (total), $M7 = 6.24\mu\text{W}$
- **Output Stage:** $M5 = 84.51\mu\text{W}$, $M6 = 96.52\mu\text{W}$

The output stage dominates the power consumption, bringing the **Total Power Dissipation to approximately $289.09\mu\text{W}$.**

4. Design 2: High-Speed Architecture

To evaluate performance scaling, a second architecture was designed with a 1:5 tail-to-output current ratio. The first stage remains unchanged, but the second stage is designed to pull $\approx 500\mu\text{A}$ to improve slew rate and naturally push the second non-dominant pole to higher frequencies.

4.1. Circuit Sizing and DC Operating Point

To handle the 5x increase in current without altering the DC node voltages, the widths of M5 and M6 were scaled proportionately.

Device	Type	W/L (μm)	Sim. I_D (μA)	$ V_{GS} $ (V)	$ V_{GS} - V_T $ (V)	g_m (μS)	r_o ($\text{k}\Omega$)
M7	NMOS	1.1/0.5	10.00	0.624	0.254	78.7	1000
M0	NMOS	13.0/0.5	100.80	0.624	0.254	793.7	100
M1, M2	NMOS	6.0/0.5	50.40	0.684	0.314	321.0	200
M3, M4	PMOS	13.0/0.5	50.40	0.685	0.295	341.7	200
M5	PMOS	128.0/0.5	505.51	0.689	0.299	3381.3	20
M6	NMOS	59.0/0.5	505.51	0.624	0.254	3980.4	20

Table 3: Simulated operating points and estimated small-signal parameters for Design 2. (*Calculated from exact LTspice node voltages; r_o assumes $\lambda \approx 0.1 \text{ V}^{-1}$*)

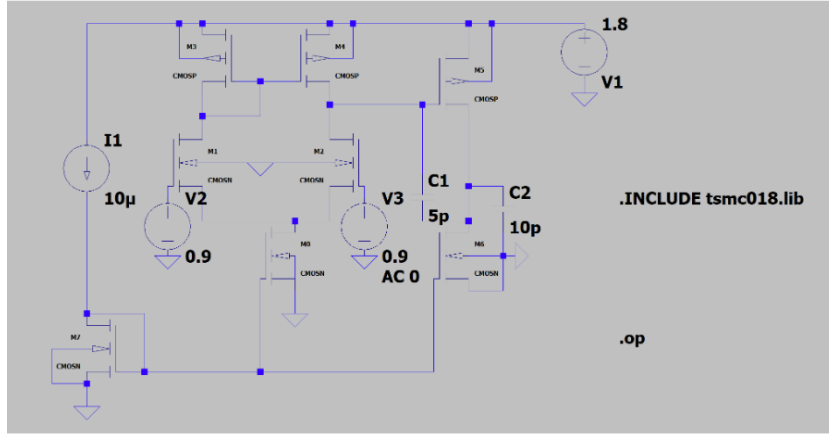


Figure 7: DC Operating Point Schematic for Design 2 ($500\mu\text{A}$ second stage).

4.2. Compensation and AC Analysis (Bode Plot)

The massive $500\mu\text{A}$ current significantly increases g_{m5} . Because the RHP zero is located at $\omega_z = g_{m5}/C_1$, this higher current naturally pushes the zero out to a much higher frequency where it no longer threatens the phase margin.

Consequently, the physical $2.5 \text{ k}\Omega$ nulling resistor was removed entirely. The circuit drives the same $C_2 = 10 \text{ pF}$ load capacitance relying solely on the 5 pF Miller capacitor for stability.

The DC gain magnitude remains virtually identical at **68.37 dB**. However, the increased drive current extends the GBW to **15.447 MHz**. Even without the nulling resistor, the circuit maintains a highly stable phase margin of **63.81°**.

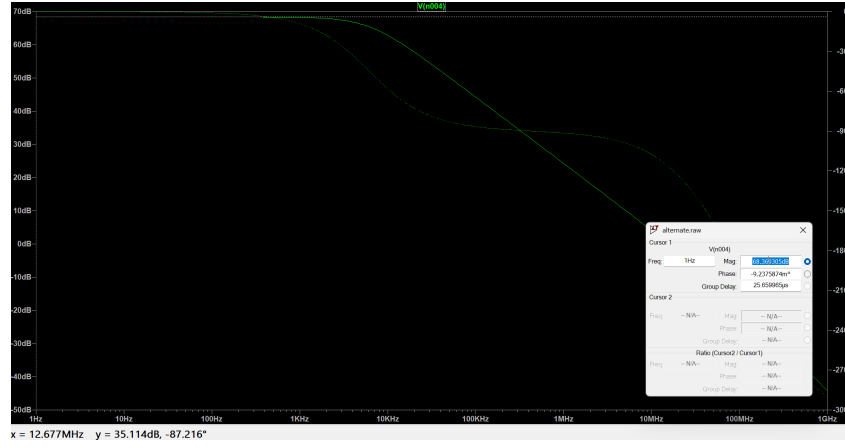


Figure 8: Bode Plot - DC gain = 68.37 dB for Design 2

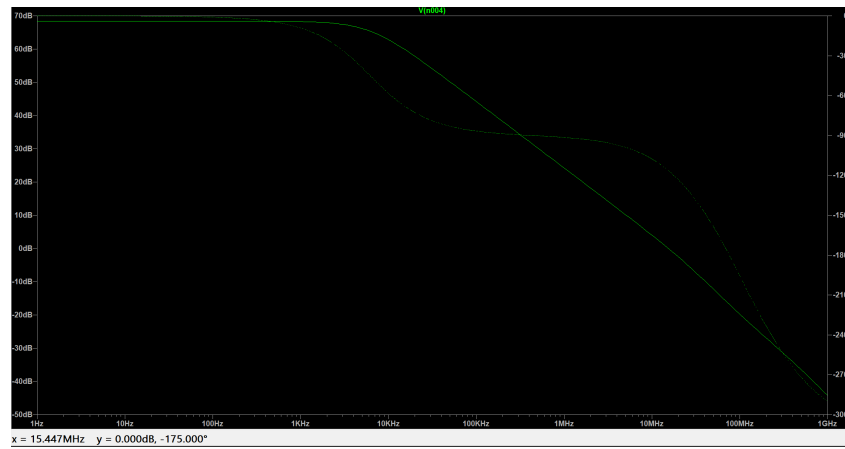


Figure 9: Bode Plot - Gain Bandwidth Product (0 dB) = 15.447 MHz for Design 2

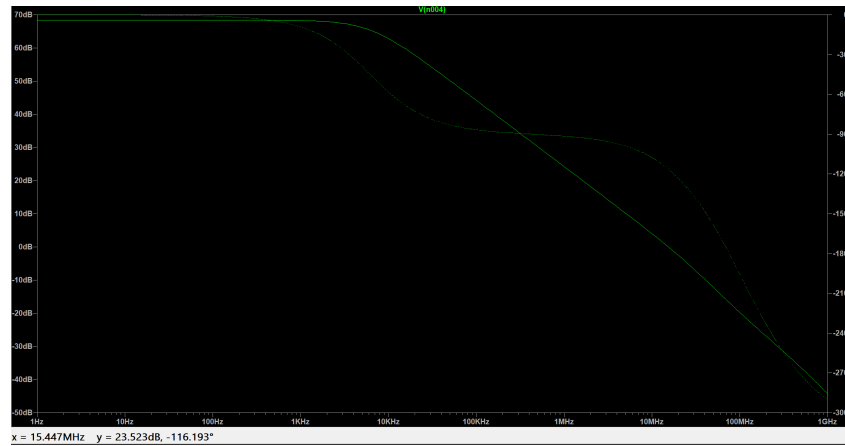


Figure 10: Bode Plot - Phase Margin = 63.81° for Design 2

4.3. Transient Response and Accuracy

Closed-loop testing was performed using the identical gain-of-2 feedback network and a 0.2V input step (configured as `PULSE(0.9 1.1 0 0.1n 0.1n 10u 20u)`). The high-drive second stage yielded an even more precise tracking response:

- **Target Output Step:** 0.4000 V
- **Measured Output Step:** 0.3997 V
- **Steady-state gain error:** 0.083%

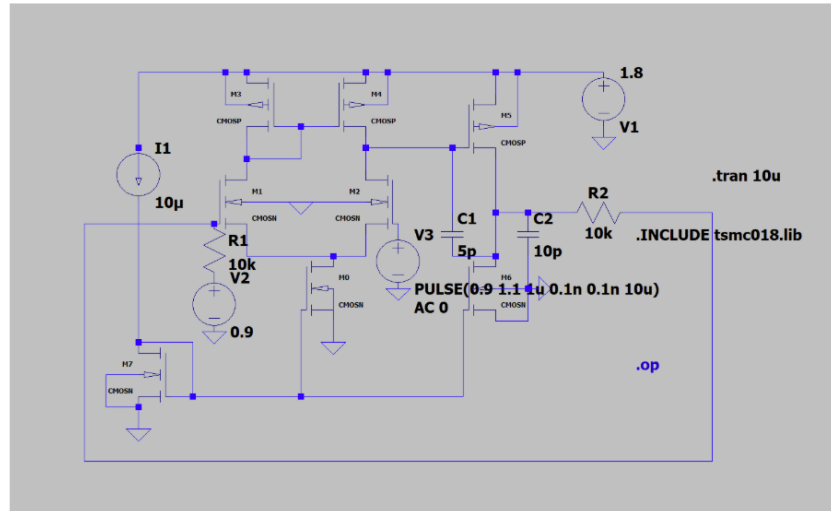


Figure 11: Closed-loop circuit - Non-Inverting Amplifier of gain = 2 using Design 2.

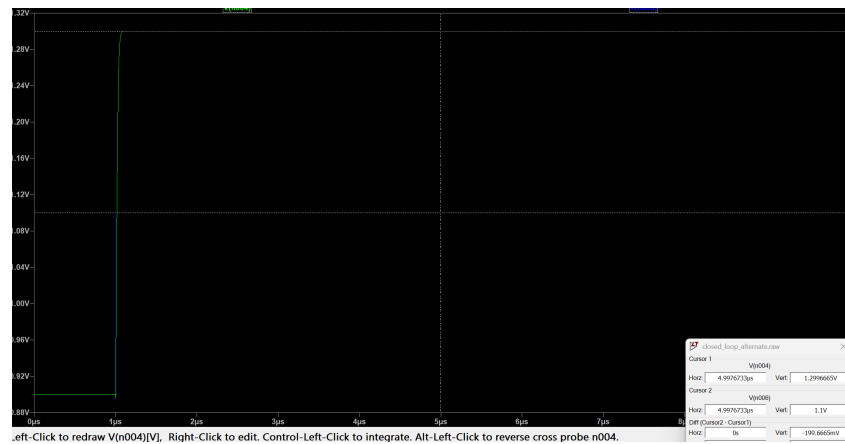


Figure 12: Closed-loop transient step response for Design 2.

4.4. Power Dissipation

The cost of the increased bandwidth and precision is severe power consumption. The per-device dissipation is:

- **First Stage & Bias:** $M0 = 22.21\mu\text{W}$, $M1/M2 = 45.17\mu\text{W}$, $M3/M4 = 34.44\mu\text{W}$, $M7 = 6.24\mu\text{W}$
- **Output Stage:** $M5 = 461.43\mu\text{W}$, $M6 = 448.49\mu\text{W}$

The total circuit power dissipation evaluates to **1.018 mW**, roughly 3.5 times higher than Design 1.

5. Conclusion and Design Trade-offs

Both OTA designs successfully met the assignment specifications, easily exceeding the 40 dB gain requirement and demonstrating stable transient tracking for an ideal closed-loop gain of 2 under a 10 pF capacitive load.

The implementation of two distinct architectures highlights a fundamental analog IC design trade-off. **Design 1** is highly power-efficient ($\sim 289\mu\text{W}$) but requires a carefully tuned series nulling resistor to combat the RHP zero introduced by its lower-current second stage. **Design 2** acts as a high-drive alternative, burning $\sim 1\text{ mW}$ of static power to achieve a higher GBW (15.4 MHz), near-zero steady-state error, and natural stability that allows for the removal of the nulling resistor entirely.

For thermally constrained applications, Design 1 is vastly superior. Design 2 is justified in scenarios where high-speed tracking and load driving are paramount.